MODULE 3: PROGRAMMING FOR BIOLOGICAL DATA

Phylogenetics

Multiple Sequence Alignment & Tree Building

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Learning Objectives

Alignment

Perform Multiple Sequence Alignment (MSA) using the msa package.

Distance

Calculate genetic distance matrices between sequences.

Trees

Build and root Neighbor-Joining trees using ape.

Outbreaks

Interpret phylogenetic clusters for outbreak investigation.

Part 1: Multiple Sequence Alignment

Finding Homology in Chaos

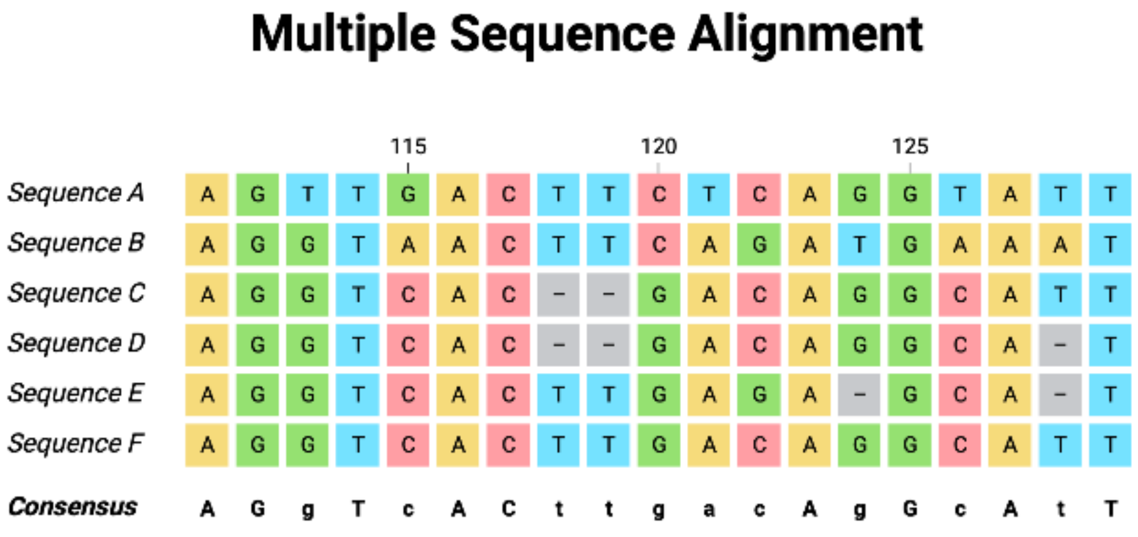
What is MSA?

Definition

Multiple Sequence Alignment (MSA) is the process of arranging three or more sequences to identify homologous regions.

Mechanism

Algorithms insert **gaps (-)** to shift sequences so that similar bases align vertically, maximizing the overall similarity score.



Why Do We Align?

Alignment is the only way to accurately compare sequences of different lengths or with insertions/deletions.

Raw (Unaligned)

```
Seq1: ATGCAT
Seq2: ATGAT
Seq3: ATGCATT
```

Hard to compare positions.

Aligned

```
Seq1: ATG-CAT-
Seq2: ATG-AT--
Seq3: ATGCATT-
```

Reveals mutations and gaps.

Popular Alignment Algorithms

Algorithm	Speed	Accuracy	Use Case
ClustalW	Slow	Good	Small datasets, classic method.
MUSCLE	Fast	Very Good	Medium datasets, general purpose.
MAFFT	Very Fast	Excellent	Large datasets (e.g., viral genomes).

The msa Package

A unified R interface for ClustalW, ClustalOmega, and MUSCLE.

Installation

```
BiocManager::install("msa")
library(msa)
```

Basic Usage

```
# Basic syntax
alignment ← msa(sequences,
                method = "ClustalW")
```

Loading Sequences

library(Biostrings) # Option 1: Read from File seqs <- readDNAStringSet("virus_seqs.fasta") # Option 2: Create manually seqs <- DNAStringSet(c("Sample1" = "ATGCATGC", "Sample2" = "ATGGATGC"))

Performing the Alignment

Running the function

```
# Align using ClustalW
my_alignment <- msa(seqs,
                    method = "ClustalW")

# Align using MUSCLE
my_alignment <- msa(seqs,
                    method = "Muscle")
```

Output Object

Returns an MsaDNAMultipleAlignment object containing the aligned sequences (with gaps).

Viewing Results

```
# Print alignment detail
print(my_alignment, show = "complete")

# Extract Consensus Sequence
msaConsensusSequence(my_alignment)
```

The print output shows sequences stacked on top of each other, making mismatches visible.

Converting for Tree Building

The msa package produces its own object type.

To build trees with the ape package, we must convert it to DNAbin format.

```
# Step 1: Convert to Biostrings
aligned_seqs <- as(my_alignment,
                   "DNAStringSet")

# Step 2: Convert to ape format
library(ape)
dna_bin <- as.DNAbin(aligned_seqs)
```

Part 2: Phylogenetic Trees

Visualizing Evolutionary Relationships

What is a Phylogenetic Tree?

A branching diagram showing the inferred evolutionary relationships among various biological species or entities.

Tips

The sequences/samples (Leaves).

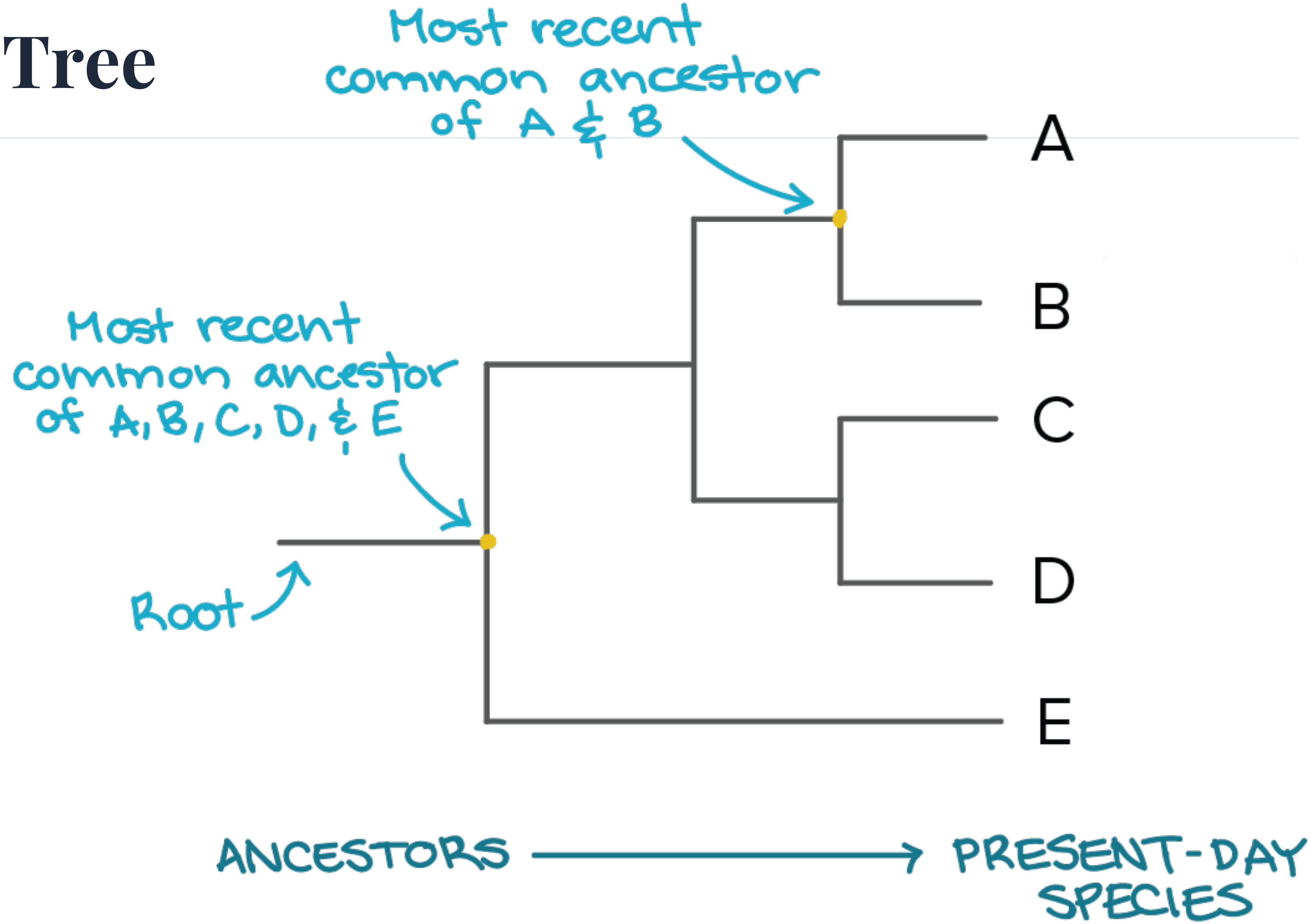
Nodes

Common ancestors (divergence points).

Branches

Length often represents genetic change.

Anatomy of a Tree



Genetic Distance

Concept

Before building a tree, we must quantify "how different" every pair of sequences is.

p-distance

The simplest measure: Proportion of sites that differ.

```
Distance = Differences / Total Sites
```

Example: 2 diffs in 10 sites = 0.20

Calculating Distance Matrix

```
library(ape)

# Calculate pairwise distances
dist_matrix <- dist.dna(dna_bin,
                        model = "raw") # p-distance

# View top corner
as.matrix(dist_matrix)[1:3, 1:3]
```

The output is a symmetric matrix where every cell [i, j] is the distance between sequence i and j.

Evolutionary Models

Evolution isn't always simple. Different models correct for hidden changes.

Model	Description
raw	Simple p-distance (observed differences).
JC69	Jukes-Cantor. Assumes equal mutation rates.
K80	Kimura 2-parameter. Distinguishes Transitions vs Transversions.
TN93	Tamura-Nei. More realistic for viral evolution.

Tree Building Methods

Neighbor-Joining (NJ)

Fast. Distance-based. Good for large initial datasets.

UPGMA

Very Fast. Assumes constant molecular clock (often unrealistic).

Maximum Likelihood

Slow. Statistical model-based. Most accurate standard method.

Neighbor-Joining Algorithm

1. Start with a star-like tree (all tips connected to center).
2. Find the pair of taxa with the shortest distance.
3. Join them into a new node (neighbors).
4. Calculate distance from new node to everything else.
5. Repeat until all taxa are joined.

Result: An unrooted tree with branch lengths proportional to change.

Building the Tree in R

1. Calculate distance dist_mat <- dist.dna(dna_bin, model = "raw") # 2. Build tree tree <- nj(dist_mat) # 3. Basic Plot plot(tree)

Rooting the Tree

Why Root?

NJ trees are unrooted (no time direction). We use an **Outgroup** (a known distant relative) to define the root.

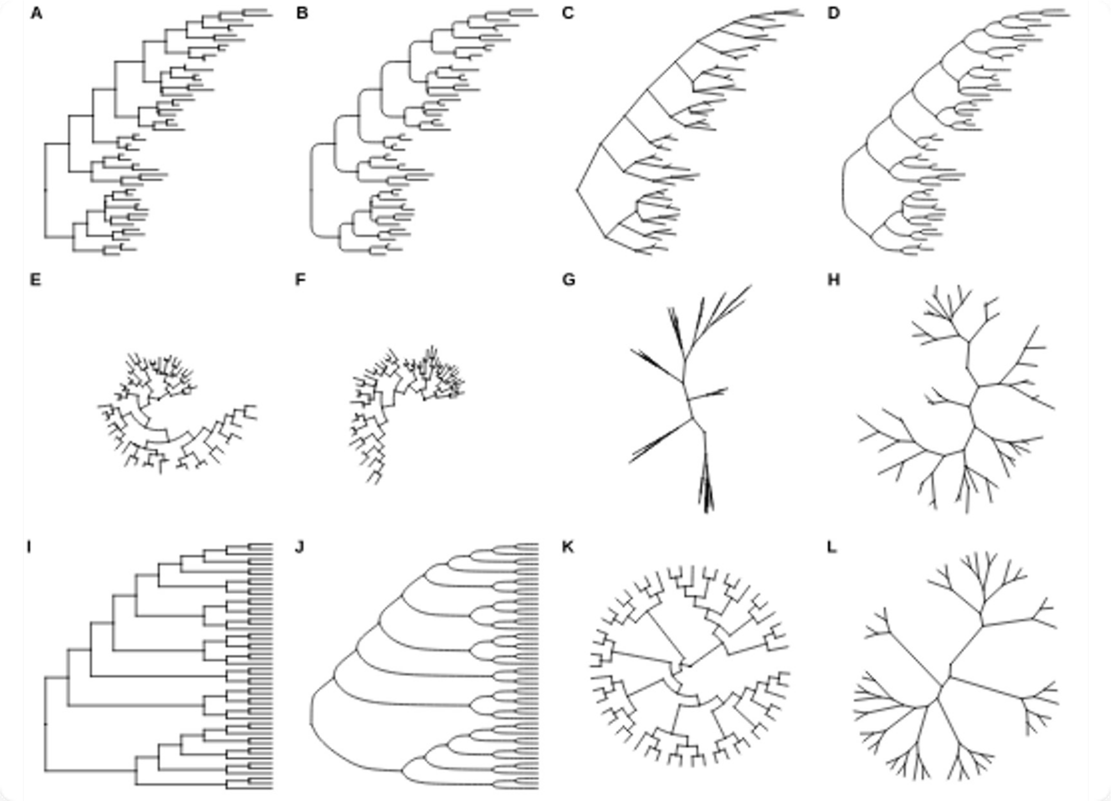
```
# Root tree using "Wuhan_Ref"
rooted_tree <- root(tree,
                    outgroup = "Wuhan_Ref",
                    resolve.root = TRUE)

plot(rooted_tree)
```

Tree Layouts

Trees can be visualized in many ways to highlight different aspects.

- **Phylogram:** Branch lengths = distance.
- **Cladogram:** Only shows branching order.
- **Fan/Radial:** Good for large trees.



Clinical Application: Outbreaks

The Question

Are these patients part of the same transmission chain?

The Logic

If patient sequences **cluster together** on the tree (monophyletic group) with very short branch lengths, they are likely epidemiologically linked.

Example: HK vs SZ Outbreak

Imagine we have patients from Hong Kong (HK) and Shenzhen (SZ).

We sequence them and build a tree with reference variants (Delta, Omicron).

- If HK patients group with Omicron...
- And SZ patients group with Delta...
- **Conclusion:** Two independent outbreaks caused by different variants.

Interpreting Clusters

```
│    ┌──── Patient 1 (Ward A)
│    │
┌────┤
│    └──── Patient 2 (Ward A)   ← Same cluster!
│
└───────── Patient 3 (Ward B)   ← Different event
```

Patients 1 & 2 are linked. Patients 3 is unrelated.

Saving Your Tree

You can save trees to open in other software (FigTree, Dendroscope).

```
# Save as Newick (Standard format)
write.tree(tree, file = "my_tree.nwk")

# Save as Nexus (Rich format)
write.nexus(tree, file = "my_tree.nex")
```

Complete Workflow

1. Load Sequences seqs <- readDNAStringSet("data.fasta") # 2. Align aln <- msa(seqs) # 3. Convert & Distance bin <- as.DNAbin(as(aln, "DNAStringSet")) dist <- dist.dna(bin) # 4. Build & Plot tree <- nj(dist) plot(tree)

Key Takeaways

- **MSA** is the foundation of phylogenetics; it identifies homologous sites.
- **Genetic Distance** matrices quantify sequence differences.
- **Neighbor-Joining (NJ)** is a fast, reliable method for initial tree building.
- **Clustering** on a tree indicates evolutionary relatedness.
- This workflow allows us to trace **pathogen transmission** paths.

Tutorial 11

Constructing the Tree

Open your R environment

Tutorial 11: Constructing the Tree

Building a Phylogenetic Tree from Outbreak Sequences

Your Scenario

The Task

You are investigating a **SARS-CoV-2 outbreak**.

Data:

- 4 Reference Variants (Wuhan, Alpha, Delta, Omicron).
- 5 Patient Samples (HK and SZ).

Goal: Assign patients to variants.

The Workflow

1	2	3	4
Align	Convert	Distance	Tree
msa()	DNAbin	dist.dna()	nj()

Steps 1 & 2: Load and Align

First, we create the dataset and align it.

```
# Load (code provided)
seqs ← DNAStringSet( ... )

# Align
# TODO: Fill in the method
alignment ← msa(seqs, method = " ... ")
```

Steps 3 & 4: Convert and Distance

Prepare the data for the ape package.

```
# Convert
dna_bin ← as.DNAbin(as(alignment, "DNAStringSet"))

# Calculate Distance Matrix
# TODO: Fill in the model
dist_matrix ← dist.dna(dna_bin, model = "...")
```

Step 5: Build Tree

Construct the Neighbor-Joining tree.

```
# Build Tree
# TODO: Fill in function
tree ← ...(dist_matrix)

# Plot
plot(tree)
```

Step 6: Rooting

Root the tree using the ancestral strain.

```
# Root
rooted_tree <- root(tree,
                    outgroup = "Wuhan_Ref",
                    resolve.root = TRUE)

plot(rooted_tree)
```

Expected Outcome

Look for Clusters

- **HK Patients (1,2,3):** Should cluster with Omicron.
- **SZ Patients (4,5):** Should cluster with Delta.

Conclusion

This suggests two separate introduction events, rather than a single cross-border transmission chain.

Session 11 Complete

You can now build phylogenetic trees in R!